CSE 150: Introduction to Computer Networks

Chris Parsa

Welcome!

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Current and Upcoming Assignments

- Reading:
  - Sections 1.1 – 1.3 of textbook
- Assignments:
  - Academic Integrity exercises and MoU: due Saturday 9/30 @11:59pm
  - Lab1: Due Saturday 10/5 @11:59pm

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Upcoming Events

- Lab1 posted on Monday
- Lab sessions start Tuesday
- Python Tutorials with student tutors next week – see schedule on Canvas

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Teaching Team

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Instructor

- Chris Parsa – please call me Chris
  - E-mail: cparsa@ucsc.edu

    No response after 24 hrs? Send it again!
  - CSE 80n and CSE 150
  - Former graduate student in the CCRG Lab
    - Advisor: JJ Garcia-Luna-Aceves
    - Research on TCP congestion control and performance over wireless links
- Office Hours: Tuesday 3-4pm in BE - 330
  - Or email for an appointment
  - Come say hi and/or ask questions!!

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Teaching Assistants and Office Hours

Lakshmi lakrishn@ucsc.edu		Wednesday 3:00pm-4:00pm Location: BE - 312B
Tyler tbmorton@ucsc.edu		Friday 2:00pm – 3:00pm Location: BE - 153A

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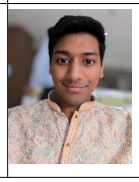
Teaching Assistants and their Office Hours

Carl cramfelt@ucsc.edu		Friday 1:00pm-2:00pm Location: BE 151
Rohit rnaguban@ucsc.edu		Monday 11:00am-12:00pm Location: BE – 153a

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Contacting us

- Please email all four of us (Chris + TAs) with questions any of us could answer
  - "CSE 150:" in subject line
  - We will Reply All
  - First to see will answer
- Contact me directly if you want to address me only

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Small Group Tutors

- Benny
- Pranav
- Brian
- Ethan

- Schedule is on Canvas
- Tutors will attend each lab session and also hold their own small group sessions

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The Class Details

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Welcome to CSE 150

- Lecture Schedule:
  - T Th 5:20PM – 6:55PM
  - Media Theater M110

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Lab Schedule

A	Tuesday	11:15am– 12:45pm	Soc Sci 1
B	Tuesday	12:45pm – 2:15pm	Soc Sci 1
C	Wednesday	1:00pm – 2:30pm	Soc Sci 1
D	Thursday	4:00pm – 5:30pm	Soc Sci 1
E	Thursday	2:45pm – 5:15pm	Soc Sci 1

You may go to other lab sessions, as often as you like to get help and work on your assignment! However, students enrolled in the lab session have priority for computer use.

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We are in person!!

(no remote option)

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Don't Come to Class if You are Sick

- If you can not attend class due to a health issue, please send me an email with the subject: attendance.
  - There is no need to provide details of your illness
- We will work together to make sure you are able to keep up with class!

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How Did you get In? Fulfill the Prerequisites

- CSE 16 (discrete math) AND CSE 12 (systems and Assembly)
- CSE 30 (Programming Abstractions: Python) or CSE 15
- Basic programming skills (project)
  - Basic Python and Unix skills are required, but can be reviewed in lab and through assignments
  - Tutoring help is available!!

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Some Specifics about CSE 150

- 7 Units
- HEAVY work load
  - Lab and Lecture
  - Minimum 14 hours/week
  - Think about your co-enrollment this quarter
- Introductory course in computer networks and protocols
  - Upper division course
  - Work and learn independently
  - No networking background required
  - Some math and calculations

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Classroom Policies

Please sit in front half of room. I need to hear you.
- Large class – 175
- Large room – 350 seats

Questions? Raise you hand! I'll finish my thought and call on you.

Electronic Devices
- For class-related use ONLY
  - If you aren't taking notes, put away

Why?
- You will learn more if focused
- You will meet your classmates
- It's only 90 minutes
- Respect for the learning environment: Phone and laptop use is DISTRACTING to those around you

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CSE 150: Course Focus

- "Fundamental knowledge" of computer networks
  - Broad introduction
  - Mainly principles, concepts and theory
  - Example network

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Why study networking?

1. Basic building block for other advanced areas of study:
   - Cyber security
   - Distributed networked systems
   - Mobile computing
   - Cloud computing
   - IoT

2. Jobs!!
   - Cisco, HP, Google, Facebook, Amazon, Microsoft, LinkedIn, Twitter, etc.

3. This class answers many interview questions!

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Where do you go from here?

- CSE 156 (winter)
  - Network Programming
- CSE 151 (spring)
  - Advanced Computer Networks
- Many graduate classes in networking

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CSE 150: Topics

- Organize devices to communicate
- How can we model a single network?
- Let's get bigger!
  - Internetworking:
- TCP/IP Protocol Stack: the Internet's standard protocol architecture used in Internet

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CSE 150: Topics (cont.)

- Network applications: fun for the end user!
  - How processes communicate
- Protocols … many many many protocols!
- Congestion control
- Routing and switching
- Link layer: wired and wireless network technologies

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We have a great example network!

How the Internet works
- Networking principles in the context of the Internet
- De-mystify the Internet by learning about the hardware, software and protocols from which it is constructed

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Learning Outcomes

Upon completion of the course, you will have a good understanding of:

- Services provided by each layer of TCP/IP Protocol Stack
- Internet architecture and common protocols
- Performance measurement in a network
- How packets are forwarded through a network
- **Steps taken and protocols employed to load a web page**

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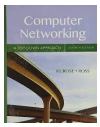
Required Textbook

Computer Networking: A Top Down Approach
by Kurose and Ross 7th or 8th edition



- Highly recommend to buy it! The book is great!
- Very helpful to complete reading assignment BEFORE every class
- >7th edition divides network chapter into 2
- On reserve in Science Library

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"Everything" is in Canvas

Communication: Students responsible to check Canvas regularly. All Announcements should be forwarded to your UCSC email.

→Receive a welcome email from me? Check Notification Settings

- Logistics Module
  - Class Syllabus – READ!!
  - Tentative 10-Week Schedule
  - Contact Us & Office Hours
  - Lab Schedule
- Weekly Modules
  - Lecture notes / slides
  - Problem Sets
- Lab Assignments
- Your Grades
- Questions? Ed Discussion on Canvas

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Lecture and Lab

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Lecture

- Attendance is required and encouraged
  - The more interactivity, the better! Listen! Ask questions!
  - Research shows students who attend class perform better and retain what they have learned
  - Warmup problem
  - Practice problems with neighbor
  - Get to know other students
- Keep up with material
  - Notes posted before class on Canvas
  - Read before class
  - Notes PDF are not a 100% replacement for lecture
  - You are responsible for **all** content presented in lecture
- Bring laptop/paper/pencil/eraser to each class

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Lab

- Supports the protocols and concepts discussed in class
  - Sometimes lecture is behind lab → Working on it!! ☺
- Each lab session:
  - TA introduces new lab – 100 points for each assignment
  - Q&A
  - Time to work on the assignments
- Attendance is expected and required
  - Progress check for current assignment (5/100 of the points)
  - If you know you can't make it, attend a different session
- Questions in lab?
  - TA plus 1 or 2 tutors in EACH lab!

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Lab (cont.)

- Software lab
  - Run some network tools over Internet on your own or lab computer
  - Unix timeshare
  - Mininet network emulator (simulator) Week 2
  - Lab exercises can be done with your laptop computer or on a lab computer
- Python programming to enhance networking concepts and support Mininet use

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Own Computer Helpful

But, Mininet VM problems:

- Student laptops Mac (M1/2)
  - See TA for support
- Mininet installed on all PCs in Soc Sci 1
  - All work can be done on lab computers

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Lab Logistics- Submission and Deadlines

- Assignments are due every Saturday @ 11:59pm
- Submission window: assignments accepted for 1 week
  - Late submissions until Friday at 11:59pm, no exceptions
  - After that, minimum score = 20 points for completed assignments
- Late Policy (within submission window):
  - 3 'grace' days for the quarter (no penalty)
    - No notification required
  - -10% per day deducted from score (starts immediately after deadline)
  - No grace days used? ….. Read syllabus! ☺

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Special note about Gradescope

See **Gradescope Submission Guidelines** on Canvas

1. Marking Requirement: Submissions on Gradescope, you *MUST* specify the page/region for each problem.
2. No Unmarked Submissions: Assignments without marked regions will *NOT* be graded and will be subject to late day deductions.

Be kind to the graders! ☺

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Student Responsibilities

- Prepare ahead of time!
  - Lab exercises are posted on Canvas on Saturday evening
  - Get familiar with the exercise before lab sessions
  - Use lab sessions to ask questions and complete assignment
- Programming Expectations
  - ✓introduction to programming class
  - Basic Python programming
    - TAs are not responsible for debugging your work
    - Logical and syntax errors should be solved on your own
    - Before you ask for debugging help, write a list of what steps you have taken to solve your problem
    - TAs will not debug over email → office hours!

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Grades / Logistics

- Course Syllabus on Canvas!
- You MUST read it

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Grading

Exams 50%

- 2 Tests: 20%
- Final Exam: 25%
  (must pass the final exam)
- Quizzes: 5%

Assignments 50 %

- Lab assignments: 30%
- Final Project: 15%
- Group Work: 5%

Approximate grading scale:

A+ [98%-100%), A [90%-97%), A- [85%-89%),

B+ [80%-84%), B [75%-79%), B- [70%-75%),

C+[65%-69%), C [60%-64%),

D [50%-59%), F (below 50%)

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Exams (50%)

- 2 Tests
- Final
- Lecture Quizzes
  - Canvas and/or paper
    - **Must** be in class, no makeups
    - Laptop or paper and sometimes in groups
    - Goal is to help students stay current with material

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Group Work (5%)

- Groups of 5 students who are in the same lab session
- Assignments due each week – usually Monday
- In person meet up 1x week!
- Purpose:
  - opportunity to discuss the course topics with your peers
  - encourage engagement with the lecture material
  - GROUP solving of problems
- If your group is not working out for some reason, please reach out to me directly
- Will assign by Tuesday

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Lab Assignments (30%)

- Course requirement to complete all labs
- 5 or 6 individual weekly lab assignments (100 points)
  - Python programming for the Mininet assignments
  - Experiments with Internet protocols
- Remember: 5 points for progress check (show to TA or tutor)

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Project (15%)

- Tentative:
  - Socket programming in Python (more info to come)
  - Project demo to TA and in-person explanation of your approach to the problem - tentative

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Help!!

- Seek help early!
- Office hours - 4 different hours in the week
- Lab
  - TAs – get to know them! Ask questions!
- Small group tutors – 4 sessions per week
- Online help:
  - Ed Discussion (online answers from teaching team)
  - Discord (student space)
- Classmates – your own network!

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Student Feedback

- Feedback is always welcome!
- Chance to do it at the end of the quarter (evals)
- But you can also do it during the quarter…
  - Come see me with any issues

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DRC Accommodations

- Submit accommodations request on new Accommodate system
- Plenty of time to complete the short quizzes
- Talk to me about your specific needs SOON! ☺

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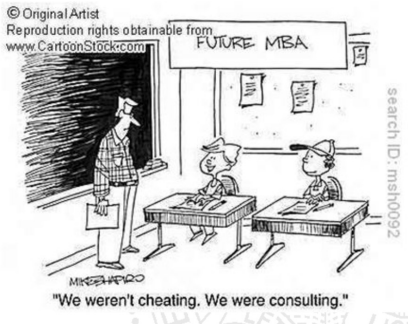
Academic Integrity



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Academic Integrity

How to Avoid Misconduct?

- All submitted work must be individual
  - Discuss ideas with classmates → submit **YOUR OWN** work.
    - All labs must be done individually
    - Project is individual
    - MOSS will be used to check your software submissions
- Quizzes must be individual (unless group quiz)
  - Any evidence of exchanging messages is an automatic FAIL
- Protect Yourself!!
  - Don't share your code
  - Don't discuss your code
  - Ask instructor and TAs if there are any questions.
  - Read Syllabus for AI policy

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New Challenges

- AI Tools
- Academic Miscondut
- Read Syllabus
- Good rule of thumb – does your answer require a citation?

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What Happens with Academic Misconduct?

- **Academic misconduct will be reported!**
  - Academic sanction: zero on assignment and may result in failing the class!
  - Disciplinary sanction: report filed to College and Engineering undergrad advising.
  - For more info, go to https://ue.ucsc.edu/academic-misconduct.html

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Academic Integrity Assignment

Work through exercises – due Monday

What is Academic Misconduct?

Do the assignment to be informed!

- Sign the Memorandum of Understanding (MoU)

Completion of this assignment REQUIRED

- No grade for future lab assignments unless this assignment is submitted

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